p8

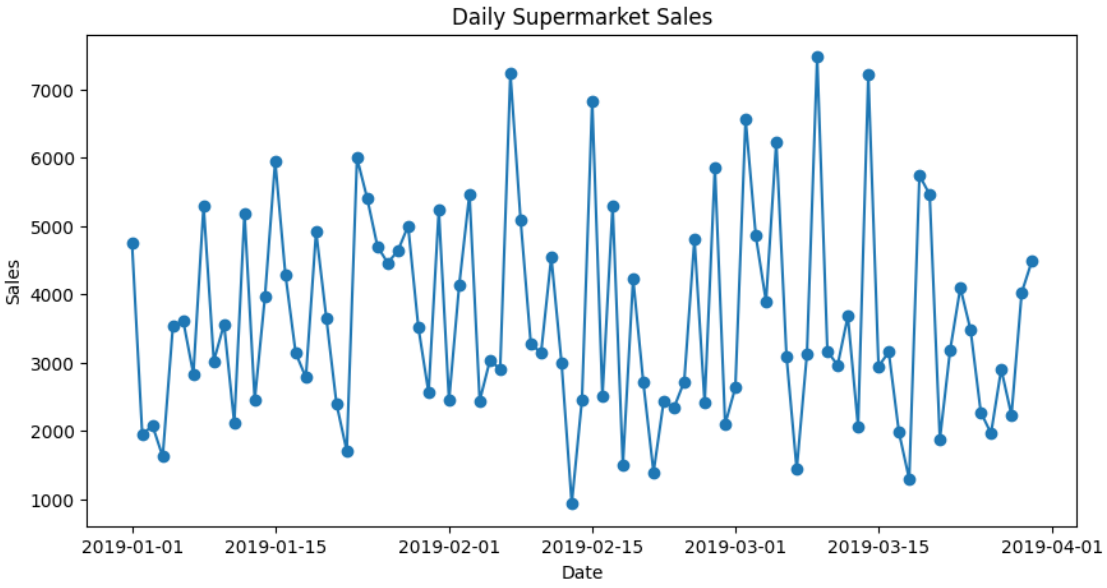
September 29, 2025

[1]:
```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.graphics.tsaplots import plot_acf
from sklearn.metrics import mean_absolute_error, mean_squared_error
import numpy as np
```

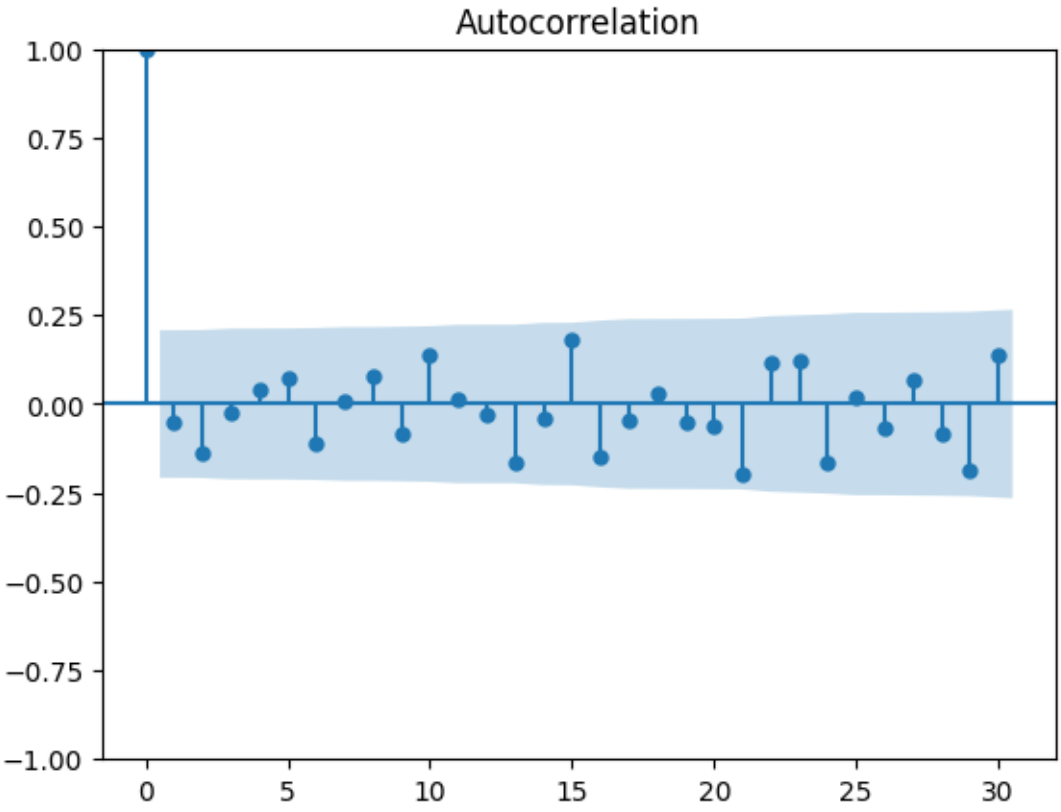
[2]:
```
# ----- STEP 1: Load dataset -----
# Assume dataset has columns: "Date" and "Total" (sales amount)
# Replace 'supermarket_sales.csv' with your actual dataset path
df = pd.read_csv("supermarket_sales.csv")
```

[3]:
```
# Convert Date to datetime and aggregate sales by day
df['Date'] = pd.to_datetime(df['Date'])
daily_sales = df.groupby('Date')['Total'].sum()

# ----- STEP 2: Plot time series -----
plt.figure(figsize=(10, 5))
plt.plot(daily_sales, marker='o', linestyle='-')
plt.title("Daily Supermarket Sales")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.show()
```



[4]:
```
# ----- STEP 3: Autocorrelation -----
plot_acf(daily_sales, lags=30)
plt.show()
```



[5]:
```
# ----- STEP 4: Train-Test Split -----
train_size = int(len(daily_sales) * 0.8)
train, test = daily_sales.iloc[:train_size], daily_sales.iloc[train_size:]
```

[6]:
```
# ----- STEP 5: Forecasting Methods -----
# Naive Forecast (last observed value)
naive_forecast = [train.iloc[-1]] * len(test)

# Average Forecast (mean of train set)
avg_forecast = [train.mean()] * len(test)

# Moving Average Forecast (window = 3 days)
window = 3
moving_avg_forecast = []
for i in range(len(test)):
    if i < window:
        moving_avg_forecast.append(train[-window:].mean())
    else:
        moving_avg_forecast.append(
            pd.concat([train, test[:i]])[-window:].mean()
        )
```

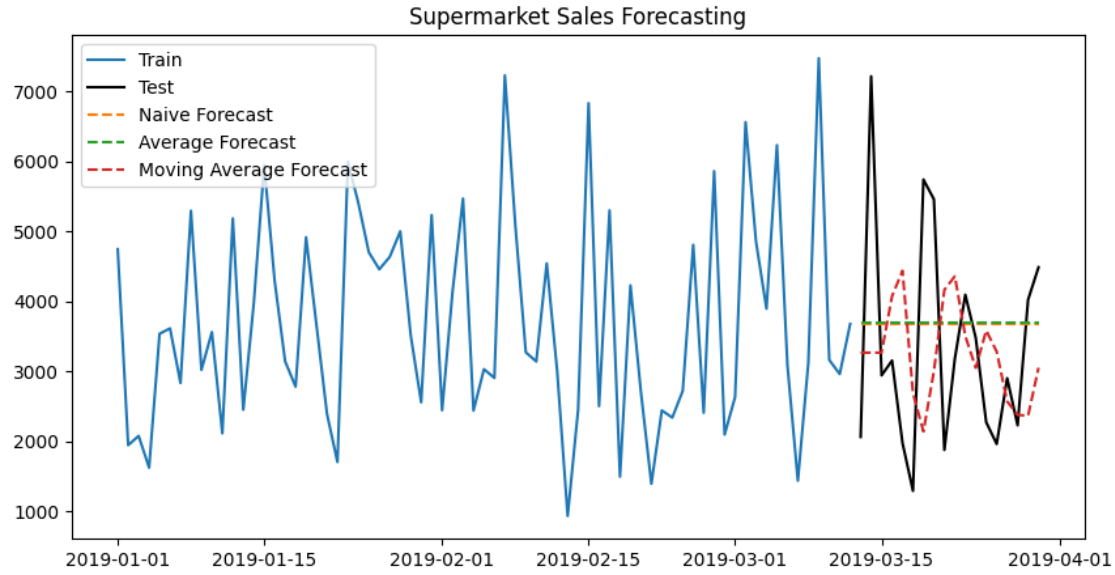
[7]:
```
# ----- STEP 6: Evaluate Accuracy -----
def evaluate(true, pred, model_name):
    mae = mean_absolute_error(true, pred)
    rmse = np.sqrt(mean_squared_error(true, pred))
    print(f"{model_name} -> MAE: {mae:.2f}, RMSE: {rmse:.2f}")

evaluate(test, naive_forecast, "Naive Forecast")
evaluate(test, avg_forecast, "Average Forecast")
evaluate(test, moving_avg_forecast, "Moving Average Forecast")
```

```
Naive Forecast -> MAE: 1319.58, RMSE: 1563.94
Average Forecast -> MAE: 1326.67, RMSE: 1568.49
Moving Average Forecast -> MAE: 1499.83, RMSE: 1833.69
```

[8]:
```
# ----- STEP 7: Plot forecasts -----
plt.figure(figsize=(10, 5))
plt.plot(train.index, train, label="Train")
plt.plot(test.index, test, label="Test", color="black")
plt.plot(test.index, naive_forecast, label="Naive Forecast", linestyle="--")
plt.plot(test.index, avg_forecast, label="Average Forecast", linestyle="--")
plt.plot(test.index, moving_avg_forecast, label="Moving Average Forecast",␣
 ↪linestyle="--")
plt.legend()
```

```
plt.title("Supermarket Sales Forecasting")
plt.show()
```



[ ]: